

Swathi Vangala

Data Engineer

 svangala89@gmail.com  +1 (317)-717-0836  Indianapolis, IN  [LinkedIn](#)

Summary

Data Engineer with around 4 years of experience in data pipeline development, data analysis, and predictive modeling using Python, SQL, MATLAB, and cloud platforms such as AWS and Azure. Proven expertise in building ETL workflows with tools like AWS Glue, Informatica, and dbt, and leveraging Hadoop (HDFS, MapReduce, Hive, and HBase) for distributed data processing. Skilled in designing interactive dashboards and visualizations in Power BI and Tableau for real-time business insights. Experienced in data warehousing on platforms like Amazon Redshift and Azure Data Lake, supporting scalable, high-performance data environments. Adept at conducting statistical analysis, A/B testing, and machine learning with tools like Apache Spark and TensorFlow for customer behavior insights and operational optimization. Well-versed in Agile methodologies and data governance practices to ensure secure, reliable data solutions for data-driven decision-making.

Skills

Programming Languages: Python (NumPy, Pandas, SciPy, SQL Alchemy, Matplotlib, PySpark), SQL, R, C++, Java

Data Warehousing: Snowflake, dbt, Amazon Redshift, Azure SQL Data Warehouse

SQL and Databases: Amazon RDS, MySQL, PostgreSQL, SQL Server, MongoDB, Cassandra

Cloud Platforms: AWS (EC2, S3, Lambda, Glue, Athena, Redshift, IAM), Microsoft Azure (Azure SQL, Azure Synapse Analytics, Data Factory, Databricks), Google Cloud Platform (BigQuery, Dataflow, Composer, Pub/Sub, Cloud SQL)

Big Data & Distributed Processing: Hadoop, Hive, Spark, Apache NiFi, Databricks

Data Visualization & BI Tools: Power BI, Tableau, Advanced Excel (Pivot Tables, VLOOKUP, Macros), Looker, AWS Quick sight.

ETL, Data Integration & CI/CD: SSIS, Informatica, Talend, Alteryx, dbt, Apache Airflow, CI/CD Pipelines, Jenkins, GitHub, Terraform

Machine Learning & Statistical Analysis: Machine Learning, Deep Learning, NLP, Computer Vision, Gen AI, Scikit-Learn

Data Modeling & Transformation: Data Modeling, Star Schema, Snowflake Schema, Normalization/Denormalization, ERwin, Lucidchart, MS Visio

Data Analysis & Processing: A/B Testing, Regression Analysis, Predictive Modeling, Hypothesis Testing, Feature Engineering, Data Mining

Project Management & Collaboration: Agile, Scrum, Waterfall, SDLC, JIRA, Confluence, Asana, MS Project

Data Governance & Security: Data Quality, Data Lineage, Data Governance Policies, AWS IAM, Azure AD

Experience

McKesson, IN - Data Engineer

Jul 2024 - Current

- Designed and maintained scalable data pipelines with Python & Scala for data ingestion, processing, and transformation.
- Implemented ETL processes through SSIS to integrate data from various systems into MySQL databases for storage and analysis.
- Built cloud-based data solutions with Azure Logic Apps & Synapse for seamless integration and processing.
- Utilized Apache Hadoop (HDFS, MapReduce, Hive, and HBase) for distributed storage and processing, optimizing scalability and performance.
- Leveraged Hadoop and Apache Spark to process and analyze large-scale datasets, applying machine learning algorithms using TensorFlow and Keras for advanced analytics.
- Employed data processing workflows by applying Pig to perform data transformations and prepare data for analysis and reporting.
- Utilized Apache Kafka for building real-time data pipelines, ensuring efficient data ingestion and processing.
- Conducted exploratory data analysis with Pandas, NumPy, Matplotlib, and Seaborn to extract insights.
- Developed interactive dashboards & reports using Tableau & SSRS to visualize key trends, facilitating data-driven decision-making.
- Designed and implemented data lake solutions using Azure Data Lake Storage (ADLS), allowing for scalable and cost-effective storage of structured and unstructured data.
- Utilized Azure Databricks for big data processing and analytics, leveraging Apache Spark for high-performance data processing.
- Implemented data pipelines using Azure Data Factory to orchestrate and automate data movement and transformation tasks.
- Adhered within Agile/Scrum methodologies to collaborate with cross-functional teams and deliver data solutions in iterative sprints.

Mathematics Assistance Center (MAC) at IUPUI - Mathematics Tutor/ Data Engineer

Jan 2024 - May 2024

- Conducted data-driven tutoring sessions for over 100 students, leveraging Python and SQL to analyze learning patterns and identify areas needing targeted instruction.
- Built a Power BI dashboard to track student progress and visualize common problem areas, enhancing reporting accuracy for academic supervisors.
- Utilized data warehousing principles with SQL Server to organize student performance metrics, enabling informed curriculum adjustments.
- Designed predictive models using Python (NumPy, Pandas, and SciPy) to forecast tutoring needs based on historical data, improving session allocation and resource planning.
- Employed data cleaning and ETL techniques in dbt to standardize and preprocess tutoring datasets, ensuring consistent and high-quality data for analysis.
- Created interactive visualizations with Power BI and Matplotlib to assist students in understanding complex mathematical concepts through data-backed teaching aids.
- Actively participated in regular team meetings to present analytical findings, optimizing tutoring strategies and improving student engagement.

Tata Consultancy Services - Assistant System Engineer/ Data Engineer

Aug 2021 - Aug 2022

- Designed and implemented data pipelines for TPG Telecom using AWS Glue, Lambda, and PySpark, streamlining data processing workflows and reducing runtime by 25%.
- Improved data warehouse performance and data product capabilities on Amazon Redshift and Amazon S3, achieving a 30% boost in data retrieval speeds for customer and network datasets.
- Developed optimized ETL processes for customer data management on MySQL and AWS Glue, reducing data inconsistencies by 20%.
- Collaborated closely with Data Product Managers and Data Architects, enhancing data accessibility and improving decision-making efficiency by 15% through automated reporting.
- Established automated Data Quality checks and defined ETL strategies, achieving a 95% success rate in data validation processes.
- Developed custom SQL scripts for data migration, data integrity checks, and performance optimization, ensuring compliance with data governance policies.
- Proficient in Azure Data Factory, Azure Synapse, and Databricks for orchestrating and processing big data workflows, ensuring high-performance data transformations and integrations.
- Documented data process workflows and system changes, improving project transparency and reducing onboarding time for new team members by 20%.
- Ensured operational efficiency and compliance of data solutions, using Amazon CloudWatch for real-time monitoring and IAM for robust access control, resulting in improved SLA adherence.
- Managed DNS routing and data accessibility through Route 53, enhancing reliability and availability of data services.
- Conducted ad-hoc data analysis with AWS Athena for deriving actionable insights from large-scale S3 datasets.
- Familiarized with Apache Airflow and NiFi for data orchestration, integrating them with AWS Glue to streamline ETL processes.
- Utilized Scrum and Agile methodologies to manage workflows, improving project timelines and delivery by 25%.

Trinity Infocom - Data Engineer

Oct 2020 - Jul 2021

- Spearheaded scalable data pipelines using Python/R for seamless ingestion, processing, & transformation.
- Employed SSMS for efficient database management, ensuring optimal storage and retrieval of data.
- Orchestrated Azure data solutions using Azure Data Factory and Logic Apps for seamless integration and processing.
- Developed ETL procedures to assure conformity, compliance with standards and lack of redundancy, translates business rules and functionality requirements into ETL procedures.
- Produced interactive dashboards and reports by applying Power BI and SSRS, enabling stakeholders to visualize key metrics and trends.
- Used Pandas for data manipulation, Sklearn for machine learning, NumPy for computations, and Matplotlib for interactive visualizations to drive data-driven strategies.
- Developed scalable data models to meet business goals, ensuring performance and maintainability. Contributed to data engineering activities such as modeling, ingestion, and manipulation to support organizational objectives.
- Crafted use-case scenarios leveraging Hive, Pig, Spark, and Spark Streaming to extract valuable insights from extensive datasets.
- Applied MapReduce techniques to discern intricate patterns within the data, fostering informed decision-making processes and propelling actionable business strategies.
- Tuned performance of slow jobs by improving design and configuration changes of PySpark jobs.

Vardhaman College of Engineering - Research Assistant/ Data Analyst

Jan 2020 – Sep 2020

- Designed and implemented statistical models to measure improvements in power efficiency and power-delay product (PDP) by 77.76% and 71.39%, respectively, compared to traditional multiplier designs, enabling data-backed optimization.
- Developed automated data pipelines in Python and Jupyter Notebooks to streamline data processing and transformation, ensuring data integrity and reproducibility.
- Performed hypothesis testing and comparative analysis between reversible and irreversible logic gates, identifying key performance indicators (KPIs) that affected multiplier efficiency.
- Used advanced data visualization techniques in MATLAB to create multi-dimensional graphs and charts, supporting detailed technical insights.
- Analyzed large datasets using Python and SQL to identify trends and patterns, supporting research in machine learning and data analytics.
- Developed statistical models to assess data quality and consistency, reducing data inaccuracies by 20%.
- Produced technical reports documenting findings and methodologies, contributing to published research in data modeling and analytics.
- Built data visualizations to simplify complex research findings, aiding stakeholders in interpreting results for strategic planning.

Education

Purdue University

Master of Science: Computational Data Science

Aug 2022- May 2024

Indianapolis, IN

Vardhaman College of Engineering

Bachelor of Technology: Electronics and Communications Engineering

Jul 2017 - Jun 2021

Hyderabad, India

Certifications and Publication

- **DP-203:** Data Engineering on Microsoft Azure. (In progress)
- **PERAM:** Ultra Power Efficient Array Multiplier Using Reversible Logic for High-Performance MAC | *ICICCT 2020*
- **QUEST:** Quantum Computing-Based Reversible Hybrid Encoder/Decoder for Error Resilient Transmission | *ICSCSP 2021*

Projects

- Detection of Crop Rows Using Advanced Image Analysis Techniques
- Integrative Deep Learning Framework for Enhanced Breast Cancer Prediction.

- Unpaired Image-to-Image Translation with CycleGAN.
- Predictive Analytics for Heart Failure Risk Assessment using Machine Learning.